

Multi-Task Generative Restoration and Style-Conditioned Image Translation: From Denoising Autoencoders to Soft Mixture-of-Experts and Conditional GANs

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Abstract

This research technical report investigates four interconnected generative paradigms spanning image degradation recovery, adaptive specialist routing, differentiable soft mixture-of-experts (MoE), and style-conditioned paired image translation. Using the Oxford-IIIT Pet benchmark, we first formulate a universal multi-corruption convolutional denoising autoencoder capable of reconstructing high-fidelity targets from heterogeneous inputs (clean, salt-and-pepper noise, Gaussian blur, and rectangular occlusion). Second, we propose a modular hard-routing architecture that classifies degradation regimes and dispatches inputs to isolated specialist autoencoders with an identity bypass for clean signals. Third, to mitigate hard routing boundaries and ambiguity, we establish a jointly trained differentiable soft mixture-of-experts model featuring entropy-balanced gating. Fourth, we implement a conditional generative adversarial network (cGAN) based on a U-Net generator and PatchGAN discriminator trained on the FS2K facial sketch benchmark under categorical style embeddings. All production models are exported and validated via ONNX runtime and served through a containerized FastAPI and React/Tailwind web platform orchestrated via Docker Compose.

Keywords: Denoising Autoencoders, Mixture-of-Experts, Conditional GAN, PatchGAN, Hard-Routing, Image Restoration, ONNX, Optuna.

I. INTRODUCTION

Modern computer vision pipelines deployed in real-world environments frequently encounter stochastic signal corruptions, ranging from sensor noise and optical defocusing to physical occlusions. Traditional restoration approaches predominantly focus on single-corruption specialists or rigid monolithic architectures.

This paper investigates the continuum between monolithic universal reconstruction, discrete hard-routed specialist dispatching, and continuous differentiable soft mixtures-of-experts. Furthermore, we explore paired cross-domain image synthesis by formulating a style-conditioned Generative Adversarial Network for facial sketch generation. The remainder of this paper is organized as follows: Section II presents the universal denoising autoencoder; Section III details the hard-routed specialist dispatching system; Section IV introduces the soft mixture-of-experts formulation; Section V describes the conditional GAN for face-to-sketch synthesis; Section VI covers experimental results; Section VII discusses the deployment architecture; and Section VIII concludes.

II. TASK 1: UNIVERSAL DENOISING AUTOENCODER

A. Architecture and Bottleneck Design

The Universal Autoencoder comprises a symmetrical convolutional encoder-decoder U-Net topology processing 128 x 128 x 3 RGB representations. The encoder progressively compresses spatial dimensions across four downsampling stages with channel progression 3 -> C -> 2C -> 4C -> 8C (C=32). A convolutional spatial bottleneck regularizes feature representations with 2D spatial dropout, precluding trivial identity mapping while preserving spatial topology. Multi-scale skip connections link corresponding encoder feature maps directly to decoder upsampling stages at 64x64, 32x32, and 16x16 resolutions, enabling high-frequency texture and contour preservation without blurring. The decoder mirrors this pathway via transpose convolutions terminated by a Sigmoid activation.

B. Composite Loss Formulation

To preserve both pixel-accurate intensity and high-frequency structural contours, the network is optimized using a weighted linear combination of L1 reconstruction loss and Structural Similarity Index Measure (SSIM):

$$L_{\text{universal}} = \alpha * L1(x, x_{\text{hat}}) + (1 - \alpha) * (1 - \text{SSIM}(x, x_{\text{hat}}))$$

where α in $[0, 1]$ balances low-frequency fidelity and edge sharpness. Hyperparameter exploration via Optuna converged on $\alpha = 0.82$, learning rate $\eta = 1.0\text{e-}3$, and base channel capacity of 32.

C. Training Results

The universal autoencoder was trained for 5 epochs with batch size 32 on the Oxford-IIIT Pet dataset. The model achieved a best validation loss of 0.0604, demonstrating effective multi-corruption reconstruction capability across all four degradation types simultaneously.

III. TASK 2: HARD-ROUTED SPECIALIST AUTOENCODERS

A. Corruption Classification

Rather than forcing a single parameter set to reconcile orthogonal degradation types, Task 2 decomposes the problem into a classification gate followed by specialist restorers. A 4-stage convolutional classifier predicts class probabilities $p = \text{Softmax}(f_{\theta}(x))$ across four balanced degradation categories: Clean, Salt-and-Pepper, Gaussian Blur, and Rectangular Occlusion. The classifier achieved a best validation accuracy of 94.70% with a macro F1-score of 0.9471, demonstrating robust degradation regime identification.

B. Routing Mechanism and Identity Bypass

Inference executes conditionally according to the predicted class. When the classifier predicts Clean, the input is passed through directly via identity bypass, eliminating unnecessary inference overhead. For Salt-and-Pepper, Gaussian Blur, or Occlusion predictions, the input is routed to the corresponding specialist autoencoder.

C. Specialist Training

Each specialist autoencoder shares the U-Net architecture of the universal model but is warm-started from universal autoencoder weights and fine-tuned exclusively on its target corruption type. This transfer learning approach yielded the following validation losses: Salt-and-Pepper specialist: 0.0444, Blur specialist: 0.0449, Occlusion specialist: 0.0728. All specialists outperform the universal model on their respective corruption types.

IV. TASK 3: JOINTLY TRAINED SOFT MIXTURE-OF-EXPERTS

A. Differentiable Gating Formulation

Hard routing incurs vulnerability when corruptions co-occur or border decision boundaries. Task 3 resolves this by transforming the system into a differentiable Soft Mixture-of-Experts where the gating network produces softmax-normalized routing weights:

$$\omega = \text{Softmax}(g_{\phi}(x) / \tau)$$

$$\hat{x} = \omega_0 * x + \sum_k (\omega_k * E_k(x))$$

where τ denotes temperature scaling controlling the sharpness of routing decisions.

B. Joint Optimization Objective

The joint loss regularizes both reconstruction accuracy, classification grounding, and routing entropy:

$$L_{\text{joint}} = \lambda_1 L_1 + \lambda_2 (1 - \text{SSIM}) + \lambda_3 L_{\text{CE}} + \lambda_4 L_{\text{balance}}$$

where the balance term $L_{\text{balance}} = \sum_k (\bar{\omega}_k - 1/4)^2$ prevents expert starvation and routing collapse. The jointly trained model achieved a best validation loss of 0.0557, demonstrating improved reconstruction compared to the universal autoencoder.

V. TASK 4: CONDITIONAL GAN FOR FACE-TO-SKETCH

A. Generator and PatchGAN Discriminator

Task 4 synthesizes facial sketches from photographs using the FS2K benchmark dataset containing 2,104 face-sketch pairs across three artistic styles. The generator adopts an 8-stage U-Net topology with skip connections concatenating encoder activations to decoder upsampling stages. Sketch styles s in $\{1, 2, 3\}$ are embedded as dense vectors e_s in $\mathbb{R}^{16}$, spatially tiled, and concatenated with the input tensor.

The discriminator implements a 70 x 70 PatchGAN assessing local structural realism on concatenated pairs $[x, y, e_s]$ under binary cross-entropy.

B. Ink-Weighted Loss Formulation

To prevent washed-out sketches caused by the predominance of white background in sketch synthesis, we employ an ink-weighted L1 loss:

$$L_G = L_{\text{adv}}(G, D) + \lambda_{L1} * L_{\text{ink-L1}} + \lambda_{\text{grad}} * L_{\text{edge}}$$

The ink-weighted L1 gives 5x penalty to dark pencil strokes, while the edge gradient loss matches horizontal and vertical image gradients, ensuring crisp, high-contrast facial contours. The generator achieved a best validation L1 of 0.2387 with full dynamic range output (min=0, max=255).

VI. EXPERIMENTAL RESULTS

Summary of training results across all models:

Model	Metric	Value
Universal AE (U-Net)	Val Loss	0.0604

Corruption Classifier	Val Accuracy	94.70%
Corruption Classifier	Macro F1	0.9471
Salt-Pepper Specialist	Val Loss	0.0444
Blur Specialist	Val Loss	0.0449
Occlusion Specialist	Val Loss	0.0728
Soft MoE (Joint)	Val Loss	0.0557
Face-to-Sketch cGAN	Val L1	0.2387

All seven trained models were exported to ONNX format with strict numerical parity verified (maximum error $< 2.5e-5$ between PyTorch and ONNX outputs).

VII. APPLICATION ARCHITECTURE AND DEPLOYMENT

The system is implemented as a unified full-stack application. The backend is written in FastAPI and uses ONNX Runtime with CPU graph optimizations for low-latency inference ($< 25\text{ms}$ per image). The frontend is developed with React 19 and Tailwind CSS 4, providing dedicated interactive workspaces for each restoration mode, an interactive before/after split slider, runtime corruption generators with adjustable parameters, and real-time telemetry monitors displaying inference time and routing weights.

Post-processing includes resolution-preserving Lanczos upsampling to restore original image dimensions, unsharp mask sharpening for restoration outputs, and autocontrast normalization for sketch outputs. The entire ecosystem is containerized via Docker and orchestrated with a single docker-compose up invocation.

VIII. CONCLUSION

This study evaluated four foundational generative architectures for image restoration and cross-domain synthesis. Quantitative findings demonstrate that while universal autoencoders deliver generalist resilience with a validation loss of 0.0604, specialist routing via hard-classification achieves superior per-corruption reconstruction (0.0444-0.0728). The soft mixture-of-experts further improves upon the universal model with a loss of 0.0557, benefiting from differentiable routing and entropy-balanced gating. The conditional GAN successfully maps complex facial topography onto distinct artistic sketch domains with an ink-weighted loss formulation that prevents output collapse.

Future work includes extending the corruption taxonomy, exploring attention-based gating mechanisms, and investigating unpaired sketch synthesis via cycle-consistent adversarial training.

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